

# MacBook Pro Keyboard Chassis

CAM Process Design & Manufacturing Engineering

## OBJECTIVE

Design the complete CNC manufacturing process for a MacBook Pro keyboard aluminum chassis using Siemens NX. From fixturing strategy through CAM programming, producing a verified In-Process Workpiece across two machining setups (top & bottom).

## MY CONTRIBUTION

**Fixture design:** vacuum workholding (bottom) + mechanical clamping (top); full 6-DOF analysis

**Tool & holder selection:** 7 Kennametal tools; custom T-cutter designed for undercut geometry

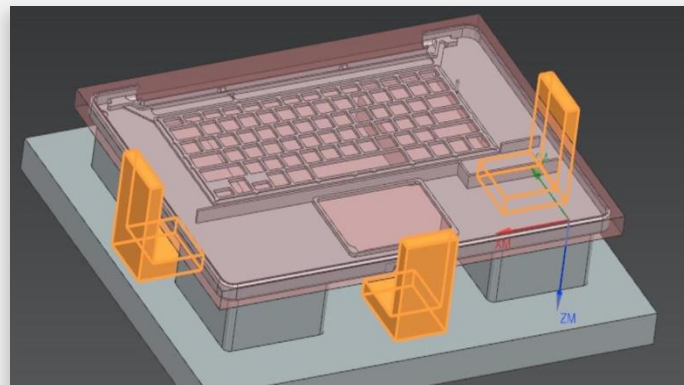
**Cutting parameters:** RPM, feed, MRR, power & torque calculated and verified within machine limits

**CAM programming:** 30 operations in Siemens NX incl. 5-axis Variable Streamline contouring

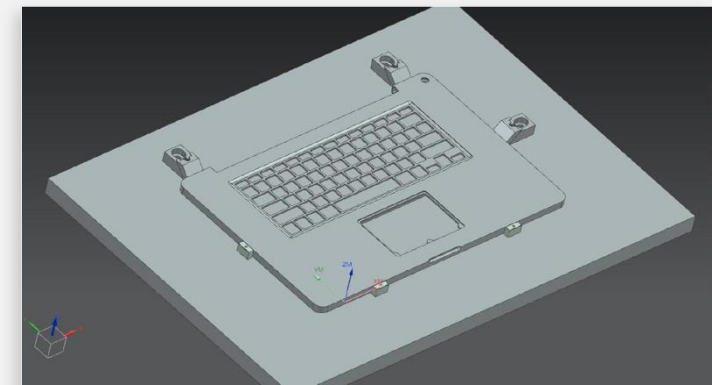
**DFM analysis:** identified 5 unmanufacturable feature classes; issued redesign recommendations

## OUTCOME

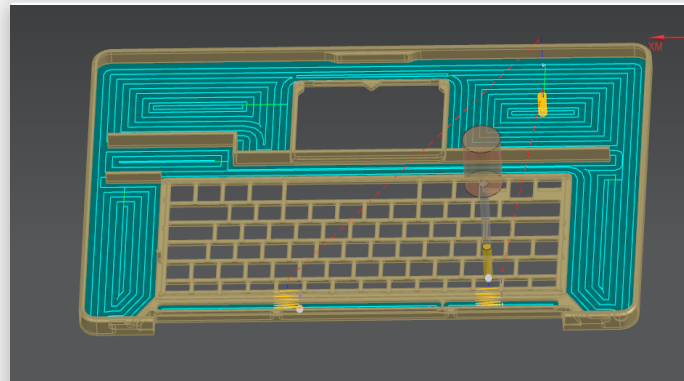
IPW verified. Zero gouges, no remaining stock, achieved  $\pm 0.05$  mm tolerance. 30 CAM operations across 2 setups. DFM feedback provided on sharp corners, tight clearances, and undercut access.



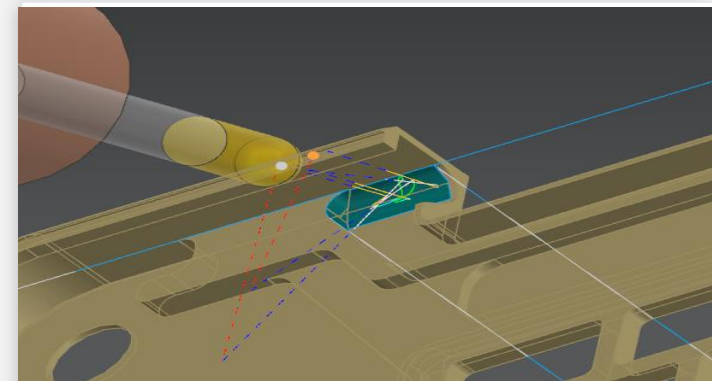
Full bottom vacuum fixture setup with removable side stops



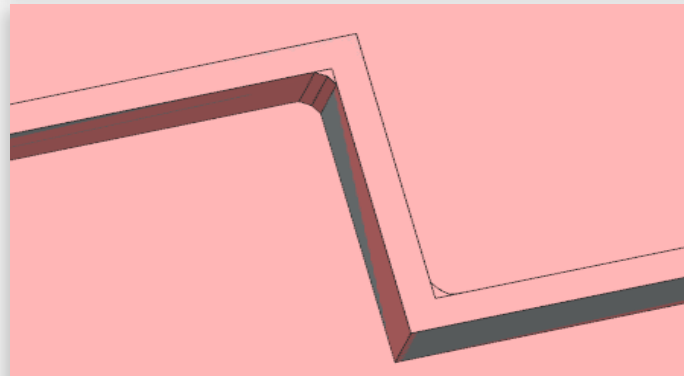
Top fixture setup with 3 low profile side clamps and 3 locating keys



Follow Periphery strategy for floor and wall



Variable streamline to cut angled grooves with 3D curvature



Unmanufacturable feature



Final IPW: bottom side (all operations complete)